

WASP-43b

R-0327

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_260222 · created 2026-07-18T03:33:38+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2461093.84 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.034
Transit depth (Rp/Rs)^2	2.01 +/- 0.95 [%]
Orbital Inclination (inc)	86.93 +/- 2.23
Airmass coefficient 1 (a1)	1.076 +/- 0.017
Airmass coefficient 2 (a2)	-0.052 +/- 0.05
Scatter in the residuals of the lightcurve fit is	1.63 %
Best Comparison Star	#2 - [391, 180]
Optimal Method	PSF photometry
Transit Duration (day)	0.0586 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.142 ± 0.034 vs 0.1594 (deviation 0.51σ)
Duration fit vs published	0.0586 d vs 0.0512 d (deviation 2.16σ)
Mid-time O–C	-6.9 min
Depth SNR	4.2
Residual scatter	1.63 %

Light curve

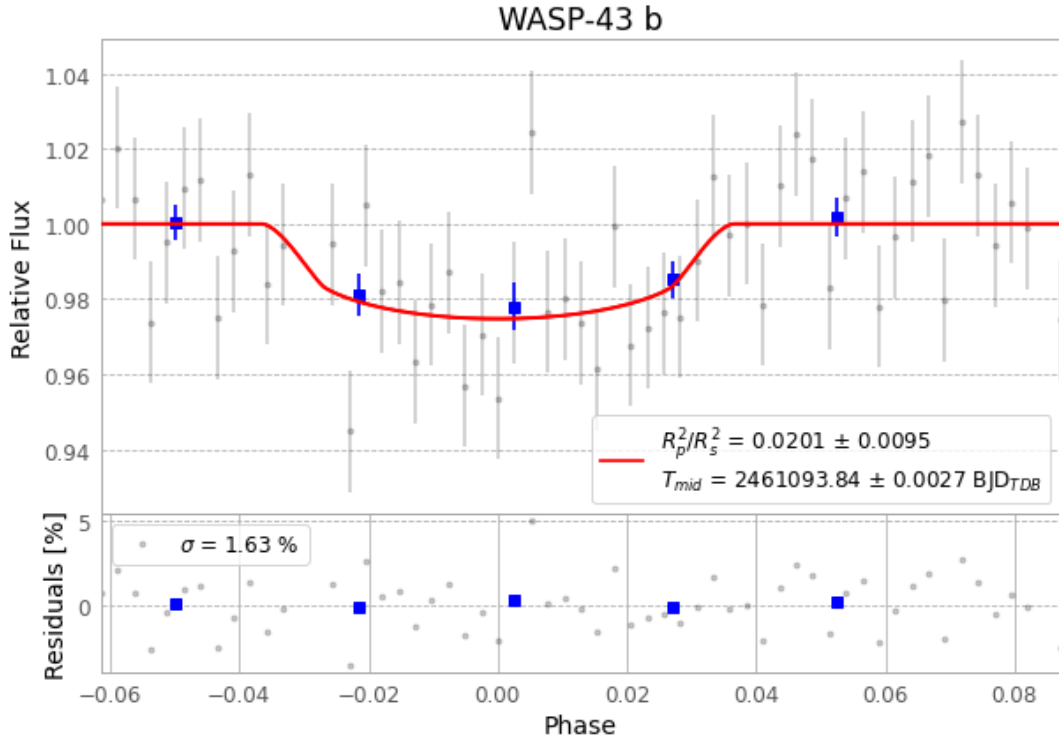


Figure 1 — FinalLightCurve_WASP-43 b_2026-02-21.png (SHA-256 39aca4d408bbb4d4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [458, 183], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9835174b1f6feb8c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

59 FITS frames (39 MB), WASP-43260222064815.FITS → WASP-43260222094216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260222.log (5ff6a7a30593...), FinalLightCurve_WASP-43 b_2026-02-21.png (39aca4d408bb...), FinalLightCurve_WASP-43 b_2026-02-21.csv (ddf9b4e39fb3...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 03:33Z.